Gödel's theorem revisited

Problem: Is it true that AIT puts limits on what mathematics can do?

In 1931, the mathematician Kurt Gödel published theorems that are often considered as the greatest revolution in mathematics ever. He showed that there are true mathematical statements that cannot receive a proof.

In 1974, Gregory Chaitin used AIT to revisit Gödel’s results and make them appear almost obvious. A mathematical theory is nothing more than a compressed version of its theorems, and this has far-reaching consequences.

Gödel's Theorem revisited through Algorithmic Information

was very naïve.

You cannot generate more computational

information than what you have.

For this reason,

the dream of an artificial

superintelligence,

that would be able to answer any

question,

is just a fantasy.

Conclusion:

The set of problems that a system

can solve is limited by the

Here we consider K as an integer function, which means that we restrict it to finite binary sequences that we interpret as integers.

Chaitin in 1974 established a result that entails Gödel’s first incompleteness theorem.

Chaitin’s theorem

Statements such as $K(n) > m$
cannot be proved above a certain value of m,
though they are true for infinitely many integers n.

The proof of Chaitin’s theorem is amazingly simple (contrary to Gödel’s historic proof).

▶ **See Chaitin s proof**

Chaitin's Incompletude theorem

0/1 point (ungraded)

Select the main argument used by Chaitin in his theorem.

- ○ Proving $K(n) > m$ makes n too simple. ✔
- ○ $K(n) > m$ cannot be true if m is large enough.

$K(n)$ is smaller than most integers m.

One can always increase n until $K(n)$ is larger than m.

✖

Answer
Incorrect:
Proving $K(n) > m$ makes n too simple, as the proof can be retrieved knowing m only. So the complexity of n becomes close to $\log m$ instead of being larger than m.

Submit You have used 1 of 1 attempt

Answers are displayed within the problem

< Previous Next >